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"Buntsandstein" magnetostratigraphy and biostratigraphic reappraisal from eastern Iberia: Early and Middle Triassic stage boundary definitions through correlation to Tethyan sections

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Abstract

A new magnetic polarity stratigraphy is reported from 214 sampling sites representing 265 m of fluviatile red beds of the Buntsandstein facies succession from the Catalan Coastal Ranges (Riera de Sant Jaume, RSJ section). The Buntsandstein constitutes the lowermost of the six lithostratigraphic units in which the Triassic from the CCR is subdivided (also grouped into the typical three-fold subdivision of the Germanic Facies from the Tethys Realm: Buntsandstein, Muschelkalk and Keuper). Magnetostratigraphic data from four sections though the uppermost Buntsandstein facies located in the Molina de Aragón area in the Iberian Ranges (Rey, D., Turner, P., Ramos, A., 1996. Palaeomagnetism and Magnetostratigraphy of the Middle Triassic in the Iberian Ranges (Central Spain). In: Morris, A., Tarling, D.R. (Eds.), Palaeomagnetism and Tectonics of the Mediterranean Region, Geol. Soc. Sp. Pub. 105, 59–82) are also discussed in the light of a new biostratigraphic reappraisal of the palynoflora content presented herein. Characteristic magnetizations are carried mostly by hematite with minor contributions by magnetite for the Buntsandstein red beds. The magnetic polarity sequence at the RSJ section consists of 9 magnetozones (and one additional less reliable magnetozone) that are represented by more than two samples. A detailed study along a magnetic reversal indicates that the nature of the remanence in the studied red beds is partially controlled by a chemical magnetization process (delayed remanence acquisition), in addition to a detrital signature (the characteristic primary direction). Chronostratigraphic constraints are provided by conodont fauna from the overlying Muschelkalk facies that indicates a middle–late Pelsonian to late Illyrian age (middle–late Anisian) (Marquez-Aliaga, A., Valenzuela-Rios, J.I., Calvet, F., Budurov, K., 2000. Middle Triassic conodonts from northeastern Spain; biostratigraphic implications. Terra Nova 12, 77–83) and a few palynostratigraphic determinations in the Buntsandstein red beds. These biostratigraphic constraints and the magnetic polarity pattern allow an unambiguous correlation of the RSJ magnetostratigraphy to the conodont-ammonoid-calibrated magnetostratigraphy from the Tethys realm (Muttoni, G., Kent, D.V., Meco, S., Balini, M., Nicora, A., Rettori, R. Gaetani, M., Krystine, L., 1998. Towards a better definition of the Middle Triassic magnetostratigraphy and biostratigraphy of the Tethyan realm. Earth Planet. Sci. Lett.

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164, 285–302; Muttoni, G., Gaetani, M., Budurov, K., Zagorchev, I., Trifonova, E., Ivanova, D., Petrounova, L., Lowrie, W., 2000. Middle Triassic paleomagnetic data from northern Bulgaria; constraints on Tethyan magnetostratigraphy and paleogeography. Palaeogeogr. Palaeoclimatol. Palaeoecol. 160, 223–237; Muttoni, G., Nicora, A., Brack, P., Kent, D.V., 2004a. Integrated Anisian–Ladinian boundary chronology. Palaeogeogr. Palaeoclimatol. Palaeoecol. 208, 85–102; Muttoni, G., Kent, D.V., Olsen, P.E., Di Stefano, P., Lowrie, W., Bernasconi, S., Hernandez, F.M., 2004b. Tethyan magnetostratigraphy from Pizzo Mondello (Sicily) and correlation to the Late Triassic Newark astrochronological polarity time scale. Geol. Soc. Amer. Bull. 116, 1043–1058). The proposed correlation identifies for the first time in the Triassic from Iberia the Olenekian (Scythian)–Anisian stage boundary (245 Ma) within magnetozone N3 in the Riera de Sant Jaume units. Likewise, the new palynostratigraphic reconsideration allows the identification of the Anisian–Ladian stage (Illyrian–Fassanian substage) boundary (taken the option at the base of the *Curionii* ammonoid Zone favored by Muttoni et al. (2004a) [Muttoni, G., Nicora, A., Brack, P., Kent, D.V., 2004. Integrated Anisian–Ladinian boundary chronology. Palaeogeogr. Palaeoclimatol. Palaeoecol. 208, 85–102] for this boundary within the upper part of the Rillo Mudstone and Sandstones Formation (RMS Formation) and the Fassanian–Longobardian substage boundary (Ladinian) within the Torete Multicoloured Mudstone and Sandstone Formation (TMMS Formation). Our data are consistent with the notion that the lower Muschelkalk transgression progressed from east to west (i.e., the Buntsandstein/Muschelkalk boundary is younger in the Iberian Ranges with respect to the Catalan Coastal Ranges).

The Early/Middle Triassic paleopole for the Catalan Coastal Ranges is located at 55.1°N 172.4E (Dp=1.4, Dm=2.7).and the Middle/Late Triassic paleopole for the Iberian Ranges is 55°N 201E (Dp=1.7, Dm=3.1). These paleopoles are compatible with the general trend of the Iberian apparent polar wander path which indicates a northward motion during the Triassic related to the general northward translation of Pangea.
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Keywords: Buntsandstein; Olenekian–Anisian boundary; Anisian–Ladinian boundary; Palynostratigraphy; Conodonts; Catalan Coastal Ranges; Iberian Ranges

1. Introduction

We present magnetostratigraphic data for the Buntsandstein facies along sections located in the Catalan Coastal Ranges (CCR) and Iberian Ranges at the eastern Iberian Peninsula. Available pollen data and other biostratigraphic constraints (conodont and ammonoid biostratigraphy) are reviewed with the aim to present an integrated chronology for the Lower and Middle Triassic succession.

The terms "Zechstein" and "Buntsandstein" are typically used as lithostratigraphic units to design mostly continental strata in Central Europe. The first is usually assigned to the Permian, whereas the Buntsandstein is the basal unit of the classic Germanic Triassic (von Alberti, 1834). The Buntsandstein is mainly clastic and was deposited in a large intracratonic basin in a fluvio-lacustrine environment, with marine influences restricted to the upper part. As a lithological facies, the Buntsandstein has no precise chronostratigraphic meaning.

The Lower Triassic reference magnetostratigraphy is chiefly based on data from the Arctic region stratotypes (Ogg and Steiner, 1991). However, significant progress has been achieved in the last decade in constructing a Triassic geomagnetic polarity sequence correlated with biostratigraphic and chronostratigraphic data in Tethyan marine sections (e.g., Gallet et al., 1992, 1993, 1994, 1998; Muttoni et al., 1994, 1995, 1996, 1997, 1998, 2004a,b). Likewise, the Germanic Triassic successions from the central European basin have been the target of thorough magnetostratigraphic studies (e.g., Nawrocki, 1997; Szurlies et al., 2003; Szurlies, 2004), as well as the onshore Triassic from Britain (e.g., Hounslow and McIntosh, 2003). Our work provides an integrated magnetostratigraphic and biostratigraphic reappraisal (palynoflora and conodont fauna) that allows a direct correlation of the Iberian sections to conodont-ammonoid-calibrated magnetostratigraphy from the Tethys realm (Muttoni et al., 1998, 2000, 2004a) and allows the recognition of both the Olenekian (Scythian)–Anisian and Anisian–Ladinian boundary (Early and Middle Triassic) in the Buntsandstein continental succession from the eastern Iberian Peninsula. This work contributes to an

improved understanding of western European inter-basin correlations during the Triassic.

2. Sections studied and methods

2.1. Catalan Coastal Ranges (CCR)

The Triassic of the Catalan Coastal Ranges (CCR) is composed of six lithostratigraphic units (Virgili, 1958; Calvet and Marzo, 1994): Buntsandstein, lower, middle and upper Muschelkalk (M1, M2 and M3 respectively), Keuper and Imón Formation. The overall thickness for the Triassic varies between 500 and 800 m. The Buntsandstein outcrops in the CCR can be grouped in three large domains: Montseny–Llobregat–Gaià, Garraf and Miramar–Prades–Priorat. These domains represent graben basins controlled by NE–SW and NW–SE trending faults, each one initially evolving independently and with imprecise palaeogeographic limits (Calvet and Marzo, 1994; López-Gómez et al., 2002 and references therein). In the Montseny–Llobregat–Gaià domain, the Buntsandstein is characterized by a sandy-siltstone composition and by a stratigraphic thickness that varies from about 150 m in the Montseny to about 300 m in the Llobregat area.

The Riera de Sant Jaume section (RSJ) (41.43°N, 1.50°W) is located in the Prelitoral Range of the CCR (Montseny–Llobregat–Gaià domain) (Figs. 1, 2) and constitutes the most complete and longest record of the Buntsandstein from that domain. At this location, the Buntsandstein facies lies unconformably over an undifferentiated Hercynian basement and is bounded at its upper part by the marine lower Muschelkalk limestones (M1). In this domain, the Buntsandstein is constituted by four lithological units starting with a basal breccia unit a few meters thick interpreted as local scree deposits. Above this unit, or directly on the Paleozoic basement, lies the Riera de Sant Jaume (RSJ) unit (up to 50 m), which comprises conglomerates, red-pink sandstones and red siltstones with

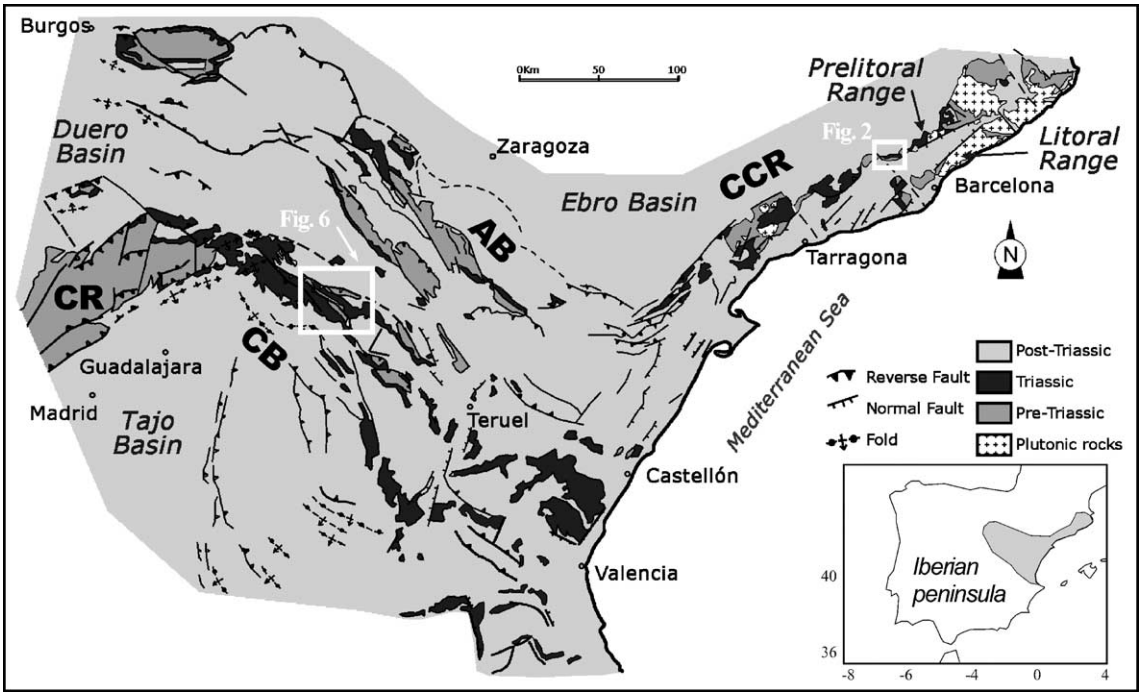


Fig. 1. Simplified geological map of eastern Iberia showing the Catalan Coastal Ranges (CCR) and the Castilian Branch of the Iberian Ranges (CB). Studied areas are framed in white and referred to corresponding figures. (CR—Central System, AB—Aragonian branch of the Iberian Ranges). Modified from Tejero and Fernández-Gianotti (2004).

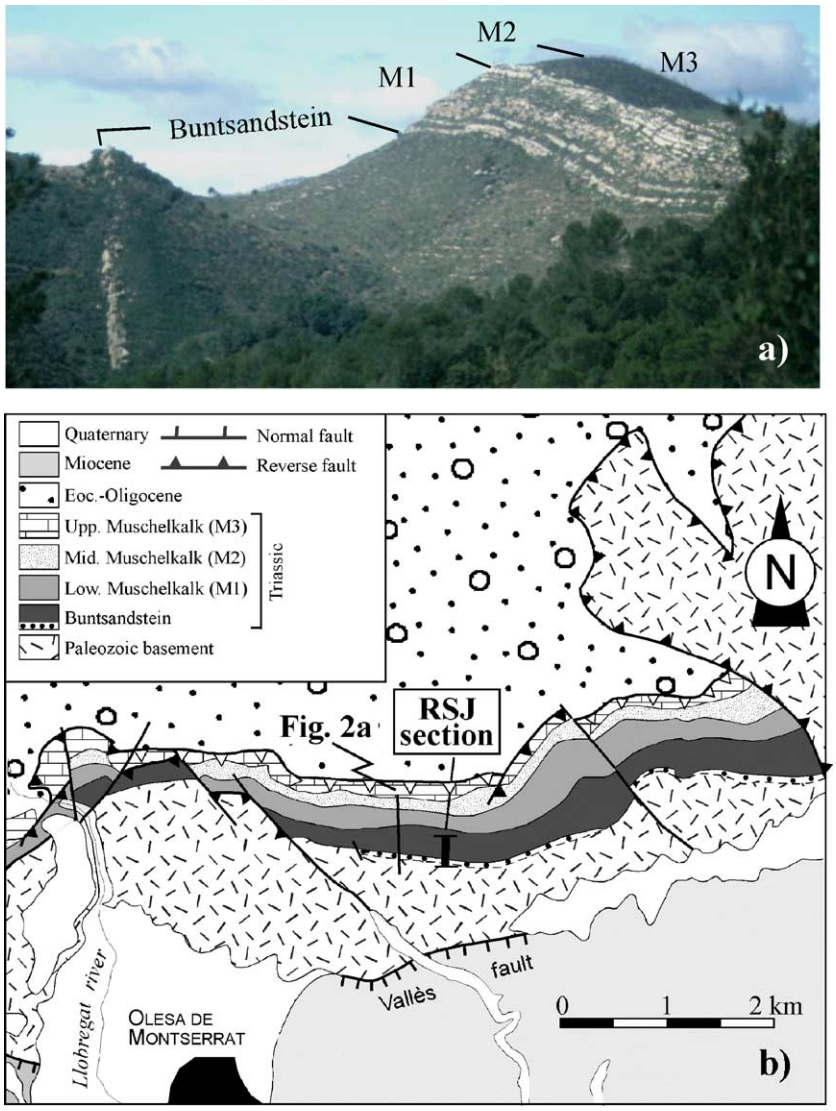


Fig. 2. Location of the Riera de Sant Jaume (RSJ) section. (a) Field picture of the Triassic succession adjacent to the RSJ section. Note the RSJ conglomeratic unit at the base of the Buntsandstein series. M1, M2 and M3 denote the lower, middle and upper Muschelkalk respectively. (b) Simplified geological map of with the location of the RSJ section and the field view shown on the top.

paleosols and represents a fining upward succession of alluvial origin (the unit is further subdivided into a lower conglomeratic subunit, an intermediate sandy subunit and an upper sandy-silty unit). The white-red pebbly sandstones of the overlying Caldes unit (up to 5 m thick) lie above the RSJ unit, or directly on the Paleozoic basement, this unit is interpreted as distal braided fluvial, mixed load sediments. The overlying Figueró unit comprises a thick (up to 235 m) succession of fining upward cycles of pink-red sandstones and red siltstones. It is interpreted as recording high sinuosity fluvial systems of mixed or suspended load. On top of the Buntsandstein succession, below the dolomitic limestones of the lower Muschelkalk, lays the upper-evaporitic–carbonatic-silty complex (Röt facies) (UECS complex), with a stratigraphic thick-

ness of about 30 m. The RSJ section represents the thickest Buntsandstein section along the CCR and the one that offers the most continuous outcrops suitable for magnetostratigraphy (Fig. 2). However, recent regional geologic mapping has suggested the presence of a thrust crosscutting the Buntsandstein series at the RSJ locality (Berástegui et al., 1996), which is difficult to identify at the RSJ section itself because of similar bedding along section and some discontinuities in the exposure (see below). Paleomagnetic sampling was performed mostly along the Riera de Sant Jaume stream but also in the adjacent road in order to minimize covered intervals. A total of 214 sites or stratigraphic levels were sampled with a portable gasoline-powered drill. The studied interval is about 265 m thick and encompasses the three lower lithologic Buntsandstein units. The uppermost 10–15 m of the Figueró unit and the upper-evaporitic–carbonatic-silty complex (about 30 m) were not sampled due to lack of suitable outcrops.

2.2. Castilian branch of the Iberian Ranges (CBIR)

The Iberian basin is currently geographically divided into the two NW–SE oriented Aragonian and Castilian branches of the Iberian Ranges (respectively AB and CB in Fig. 1), separated by the Tertiary Calatayud–Teruel basin. Permian and Triassic sediments were deposited in a series of intramontane, transtensional, half-graben basins and the stratigraphic succession can vary substantially in thickness from hundreds to thousands of meters owing to varied subsidence rates (Sopeña et al., 1988; López-Gómez et al., 2002 and references therein). A number of stratigraphic units have been described for the Permo-Triassic mostly continental sequences of the Iberian Cordillera which are associated with the three discrete lithologic facies characteristic of the so-called Germanic Triassic, namely Buntsandstein (siliciclastic), Muschelkalk (carbonate) and Keuper (evaporitic). In the Molina de Aragón study area in the central part of the Castilian branch, the Buntsandstein sediments unconformably overlie all previous rocks and represent a period of major subsidence with accumulation of some 550 m of clastic red bed sequences. The two concerned units in this work, the Rillo Mudstone and Sandstones Formation (RMS Formation) and the Torete Multicoloured Mudstone and Sandstone Formation (TMMS Formation), represent the evolution from a distal fluvial environment to the supratidal sedimentation which preceded the Tethys transgression, which in this area is marked by the overlying Tramacastilla Dolomites Formation (TD Formation). The RMS Formation comprises up to 120 m of red mudstones and sandstones with local basal conglomerates. The overlying TMMS Formation (40 m) consists of thin-bedded mudstones and sandstones, which often show evidence of evaporite precipitation and occasional interbedding of fine dolomitic horizons showing algal lamination. Five sections of these formations have been examined earlier in a paleomagnetic and magnetostratigraphic study (Rey et al., 1996). The RMS Formation was sampled in the Rillo de Gallo section, while the TMMS Formation was sampled in four separate sections, namely: La Hoz del Gallo; Aragoncillo; Ermita de la Virgen del Buen Labrador and Arroyo de San Roman.

3. Paleomagnetic methods

A gasoline-powered drill was used for paleomagnetic sampling. Cores (2.54 cm in diameter) were cut in the laboratory into standard specimens (2.2 cm in length) for paleomagnetic measurements. Natural remanent magnetization (NRM) and remanence at all demagnetization steps were measured using a 2G-Enterprises cryogenic magnetometer equipped with DC-squids at the Ludwig–Maximilians-Universität München paleomagnetics laboratory (RSJ section) and the cryogenic magnetometer at Close House in Newcastle for the Iberian Range samples. Standard shielded furnaces were used for thermal demagnetization.

Usually one or two specimens per stratigraphic level were routinely subjected to stepwise thermal demagnetization. Thermal demagnetization up to 690 °C included a maximum of 13 steps with intervals of 20–100 °C. Characteristic remanent magnetizations (ChRM) were computed by least-squares fitting (Kirschvink, 1980) on the orthogonal demagnetization plots (Zijderveld, 1967). The mean ChRM declination and inclination for each sample have been used to derive the latitude of the virtual geomagnetic pole (VGP). This parameter has been used as an

indicator of the polarity (normal polarity for positive VGP latitudes and reverse polarity for negative VGP latitudes).

4. Biostratigraphic review

The available palynologic data for the Buntsandstein in eastern Iberia have been reinterpreted following the methodological approach outlined in the PhD thesis of one of us (Diez, 2000), in which a new Triassic palynostratigraphic scale for the Western Peritethyan Domain is developed. In that scheme, valid taxa are only defined by clearly identified species, disregarding the ones that have been referred with the terminology *confer* (cf.) or *affinis* (aff.). The "absent criteria" to determine pollen associations are also not used given that the actual number of well-defined taxa allows basing the dating on registered entities only.

4.1. Catalan Coastal Ranges (CCR)

There are few palynologic data for the Buntsandstein facies from the Triassic succession in the Catalan Coastal Ranges. The scarcity may be related to the important biological crisis following the Permian–Triassic boundary but is most likely due to bad preservation conditions (Diez et al., this volume). There are only three studies available from the CCR Buntsandstein facies that we will evaluate.

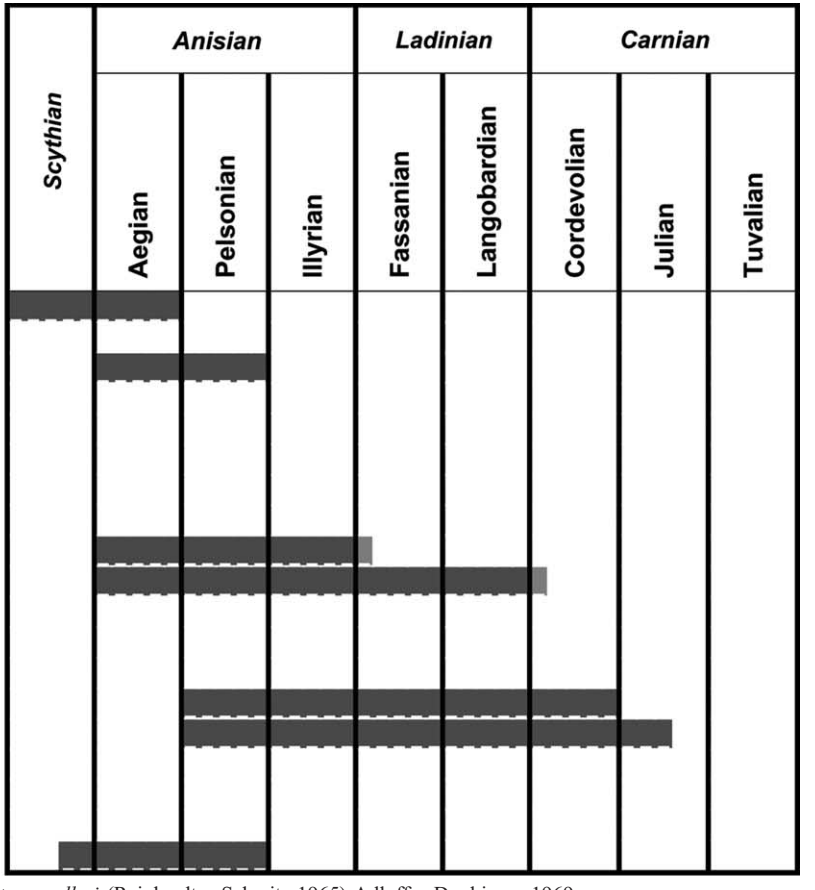
Visscher (1967) reports a pollen assemblage from a grey argillaceous level within the sandy Buntsandstein conglomerates in the vicinity of Montroig del Camp (Tarragona) (Prades–Colldejou sector, Gómez-Gras, 1993; Miramar–Prades–Priorat domain, Marzo, 1980), which is considered similar to the German Röt. However, the presence of *Alisporites grauvogeli*, *Voltziaceaesporites heteromorpha* and *Angustisulcites gorpii* could indicate an Aegean (early Anisian) age. In any case, the imprecise stratigraphic location of the fossiliferous levels in the original publication makes the data less relevant although we can presume that correspond to the youngest Buntsandstein levels. The "upper-evaporitic–carbonatic-silty complex" (Röt facies) of the Buntsandstein succession in the Hostalets de Balenyà (Montseny sector, Gómez-Gras, 1993; Montseny–Llobregat domain, Marzo, 1980) studied by Calvet and Marzo (1994) and in the la Pena locality (L'Espluga de Francolí, Miramar–Prades–Priorat domain, Marzo, 1980) studied by Solé de Porta et al. (1987) is assigned to the Anisian (most likely early Anisian) on the basis of the presence of *Stellatopollenites thiergartii* (=*Hexasaccites muelleri*), *Voltziaceaesporites heteromorpha*, *Triadispora* sp. and the absence of *Praecirculina granifer*. The appearance of *Alisporites grauvogeli* and the different *Triadispora* and *Microcachryidites fastidioides* species would confirm an age near to the early–middle Anisian transition close to the Aegean–Pelsonian boundary. In our opinion, the absence criteria, in this case *Praecirculina granifer*, should not be used to date a paleontologic level.

The palynologic data available for the Buntsandstein facies from the CCR are shown schematically in Tables 1 and 2. The RSJ magnetostratigraphic section is located in the Prelitoral Range in the Llobregat area (Montseny–Llobregat–Gaià domain), and therefore, the palynologic data from the Hostalets de Balenyà locality from the same domain are a first-order chronological constraint. The La Pena locality is situated in the Prades area (Miramar–Prades–Priorat domain) while the data from Visscher (1967) on the Prades–Colldejou sector (Miramar–Prades–Priorat domain) are not considered due to the imprecise stratigraphic position of the levels studied.

Further chronostratigraphic constraints for the Buntsandstein in the CCR are provided by biostratigraphic determinations in the overlying marine lower Muschelkalk (M1) that is divided into four members, which are, from base to top, El Brull, Olesa de Montserrat, Vilella Baixa and Colldejou (Calvet and Marzo, 1994 and references therein) with an overall stratigraphic thickness of about 70 m. The lower Muschelkalk has classically been ascribed to the Anisian on the basis of the presence of *Spiriferina (Mentzelia) mentzeli* (Virgili, 1958) at the base of the unit as indication of the lowermost Anisian (the base Anisian being in the underlying Buntsandstein). The presence of a *Paraceratites* fauna in the middle part of the lower Muschelkalk appears to be correlatable with the *binodosus* and *trinodosus* levels of the late Anisian (Virgili et al., 1977; Goy, 1995). A coherent conodont-chronostratigraphic estimate for the lower Muschelkalk facies has recently been established by

Table 1
Palynological assemblages for the Buntsandstein facies in the Catalan Coastal Ranges

Palynological assemblages	Visscher (1967)	Calvet and Marzo (1984)	Solé de Porta et al. (1987)
Alisporites grauvogeli Klaus 1964	x	x	x
Alisporites microreticulatus Reinhardt 1964			x
Angustisulcites gorpii Visscher 1966	x		
Angustisulcites klausii Freudenthal 1964	x		
Apiculatasporites plicatus Visscher 1966	x		
Cyclogranisporites arenosus Mädler 1964			x
Falcisporites snopkovae Visscher 1966			x
Kraeuselisporites hoofddijkensis Visscher 1966	x		
(*) *Stellatopollenites thiergartii* (Mädler) Clement-Westerhof *et al.* 1974		x	
Microcachryidites fastidioides (Jansonius) Klaus 1964			x
Striatoabieites aytugii (Visscher) Scheuring 1978		x	x
Taeniaepollenites(Lunatisporites) jonkeri Visscher 1966	x		
Taeniaepollenites (Lunatisporites) multiplex Visscher 1966	x		
Triadispora crassa Klaus 1964			x
Triadispora falcata Klaus 1964			x
Triadispora plicata Klaus 1964			x
Tubantiapollenites balmei (Klaus 1964)Visscher 1966	x		
Verrucosisporites jenensis Reinhardt & Schmitz in Reinhardt 1964	x		
Voltziaceaesporites heteromorpha Klaus 1964	x		x



(*) *Stellatopollenites thiergatii* (Mödler) Clement-Westerhof et al. 1974 = *Hexasaccites muelleri* (Reinhardt y Schmitz 1965) Adloff y Doubinger 1969.

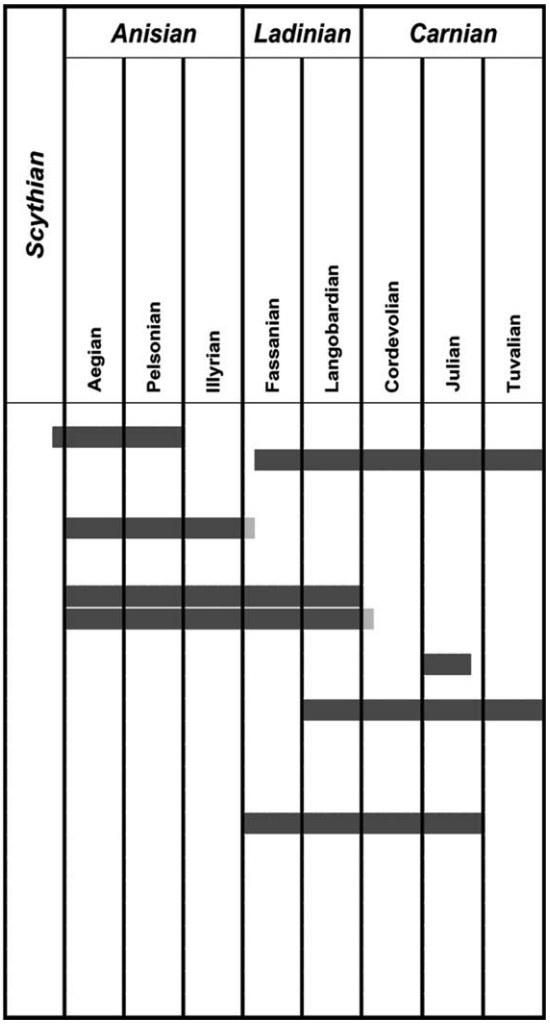
Table 2
Proposed age of available pollen determinations for the Catalan Coastal Ranges

Facies	Stratigraphic unit	Locality	Original age determination	Proposed age Diez (2000)
Buntsandstein	Upper-evaporitic–carbonatic-silty complex (UECS complex)	Els Hostalets de Balanya Calvet and Marzo (1994) La Pena (L'Espulga del Francoli) Solé de Porta et al. (1987)	Aegenian–Pelsonian	Late Aegenian–early Pelsonian

The "Proposed age" considers all the individual determinations as reinterpreted by Diez (2000) and herein. Dashed lines indicate an approximate position.

Table 3
Palynological assemblages for the Rillo Mudstone and Sandstones Formation (RMS Formation) and the Torete Multicoloured Mudstone and Sandstone Formation (TMMS Formation), in the Iberian Ranges

Palynological assemblages	L.A.R. Fm.			L.A.A.T. Fm.		
	Ramos (1979)		Sopeña et al (1995)	Pérez-Arlucea (1986)	Sopeña et al (1995)	Ramos (1979)
	Riba de Saelices Lev. 14 Rillo de Gallo Lev. 149	Rillo de Gallo Lev. 241	Luzón Lev. 36	Castellar de la Muela Lev. 15 Nebouera Lev. 13 &15	Luzón Lev. 37	Riba de Saelices
Alisporites cordiformis (Jansonius) Klaus 1964	x					
Calamospora tener (Leschik *in* Kräusel & Leschik 1956) Mädler 1964				x		
Camerosporites secatus Leschik 1956						x
Duplicisporites granulatus (Leschik 1956) Scheuring 1978		x	x		x	x
Enzonalasporites sp. div.	x	x	x		x	
Hexasaccites muelleri (Reinhardt & Schmitz 1965) Adloff & Doubinger 1969	x					
Jugasporites renalis (Klaus) Grauvogel-Stamm	x					
Keuperisporites baculatus Schulz 1965						x
Microcachryidites doubingeri Klaus 1964				x		
Microcachryidites fastidioides (Jansonius 1962) Klaus 1964	x	x	x		x	x
Ovalipollis pseudoalatus (Thiergart) Schuurman 1976		x	x		x	x
Paracirculina maljawkinae Klaus 1960						x
Paracirculina scurrilis Scheuring 1970						x
Paracirculina tenebrosa Scheuring 1970						x
Praecirculina granifer (Leschik *in* Kräusel & Leschik 1956) Klaus 1960				x		
Protodiploxypinus gracilis Scheuring 1970				x		
Striatoabieites aytugii (Visscher) Scheuring 1978		x	x		x	
Taeniaepollenites (lunatisporites) jonkeri Visscher 1966		x	x		x	
Triadispora aurea Scheuring 1970		x	x		x	x
Triadispora crassa Klaus 1964				x		
Triadispora epigona Klaus 1964				x		
Triadispora falcata Klaus 1964	x	x	x	x	x	
Triadispora plicata Klaus 1964				x		x
Triadispora staplinii (Jansonius 1962) Klaus 1964	x	x	x	x	x	
Triadispora suspecta Scheuring 1970		x	x		x	
Triadispora vilis Scheuring 1970		x	x		x	
Verrucosisporites remyanus (Mädler) Adloff & Doubinger 1969	x					



Marquez-Aliaga et al. (2000). Two sections were studied: the Ametlla section from the Miramar–Prades–Priorat domain, and the Riera de Sant Jaume section from the Montseny–Llobregat–Gaià domain. Recognized conodont taxa include *Paragondolella bulgarica*, *P. hanbulogi*, *P. bifurcata*, *Neogondolella constricta*, *N. cornuta*, *N. excentrica* and *N. basisymetrica*. The conodont sequence allows recognition of the *P. bulgarica* and *N. constricta* zones, which indicate a middle–late Pelsonian to late Illyrian age (middle–late Anisian). The base of the *N. constricta* zone is located within the Olesa de Montserrat member at about 10.5 m from the base of the lower Muschelkalk series.

4.2. Castilian branch of the Iberian Ranges (CBIR)

According to our new interpretational approach outlined above, the published palynological analysis for the Triassic Rillo Mudstone and Sandstones Formation (RMS Formation) and the Triassic Torete Multicoloured Mudstone and Sandstone Formation (TMMS Formation) from the Castilian branch of the Iberian Ranges can be reassessed. We will make a chronological inventory of data as originally published and review it with the new criteria.

In Ramos (1979), level 14 from the Riba de Saelices section and level 149 from de Rillo Gallo section, both are given a late Anisian–early Ladinian age in the RMS formation. However, the presence of *Hexasaccites muelleri* indicates Anisian while the appearance of *Triadispora falcata* would further constrain the age to be Pelsonian–Illyrian as suggested by Diez (2000).

Pérez-Arlucea (1986) assigns an Anisian–Ladinian age to the TMMS formation based on the presence of *Calamospora tener*, *Triadispora staplinii*, *T. falcata*, *T. suspecta* and *Alisporites* sp. in samples from the Muela (level 15) and Noguera (levels 13 and 15) localities. Pérez-Arlucea (1986) notes that such an assemblage is not conclusive but a late Anisian or early Ladinian age is inferred considering the similarities with the assemblage described in Ramos (1979). This seems to be logical given that the successions studied are placed in equivalent stratigraphic positions. As suggested by Diez (2000), the *Protodiploxypinus gracilis*, *Triadispora falcata* and *Microcachryidites doubingeri* assemblage would indicate a large temporal span from the Pelsonian to the Langobardian substages.

A synthesis of the palynological studies from the Castilian branch of the Iberian Ranges is given in Sopeña et al. (1995). A synthetic composite assemblage that includes the upper part of the RMS formation (level 36 from the Luzón section and level 241 from the Rio de Gallo section) and the base of the TMMS formation (level 37 from Luzón) (Ramos, 1979) is given. The presence of *Triadispora aurea* and *T. suspecta* in the assemblage, in addition to

Table 4
Proposed age of available pollen determinations from the northern sector of the Castilian branch of the Iberian Ranges

Facies	Formation	Locality and level	Original publication	Original age determination	DIEZ (2000)	
					Reinterpreted age	Proposed age
Buntsandstein	Torete Multi-coloured Mudstone and Sandstone Formation (TMMS)	Riba de Saelices	RAMOS (1979)	Early Carnian (*)	Late Fassanian Early Julian.	Late Fassanian
		Luzón (lev. 37)	SOPEÑA et al. (1995)	Ladinian sensu lato	Ladinian sensu lato	
		Castellar de la Muela (lev. 15) Noguera (lev. 13 and 15)	PÉREZ-ARLUCEA (1986)	Anisian-Ladinian (Illirian-Fassanian?)	Pelsonian-Langobardian	
	Rillo Mudstone and Sandstones Formation (RMS)	Luzón (lev. 36)	SOPEÑA et al. (1995)	Ladinian sensu lato	Ladinian sensu lato	Early Fassanian
		Rillo de Gallo (lev. 241)	RAMOS (1979)	Ladinian sensu lato	Ladinian sensu lato	
		Riba de Saelices (lev.. 14) Rillo de Gallo (lev..149)	RAMOS (1979)	Illirian-Fassanian	Pelsonian-Illirian	Illirian / Pelsonian

The "Proposed age" considers all the individual determinations as reinterpreted by Diez (2000) and herein. Dashed lines indicate an approximate position.

several *Ovalipollis* species, is taken as indication of Ladinian sensu lato. Diez (2000) agrees with the age interpretation but indicates that the age should be based upon the presence of *Triadispora aurea* and *Microcachryidites fastidioides* instead, and not *Triadispora suspecta* that has in principle a broad presence range.

Ramos (1979) reports a pollen assemblage in the TMMS section from Riba de Saelices as early Carnian. Diez (2000) indicates an age between the late Fassanian and the early Julian considering the overlap of the *Keuperisporites baculatus* and *Camerosporites secatus* biozones.

All the palynologic revisions outlined above together with the original interpretations are illustrated in a common stratigraphic framework in Tables 3 and 4.

5. Magnetostratigraphy

5.1. Riera de Sant Jaume (RSJ) section

A total of 282 specimens were stepwise thermally demagnetized (AF demagnetization proved unsuitable due to the presence of high coercivity minerals). The intensity of the NRM varied in the range 2–8×10^{-3} A/m. A ChRM component trending toward the origin of the demagnetization diagrams is usually isolated at temperatures above 630 °C (often 650 °C) up to 690 °C (Fig. 3). These high temperatures are consistent with detrital hematite or ilmenohematite mineral phases as remanence carriers. The ChRM has either normal (Fig. 3c and i) or reverse polarity (Fig. 3a and b) in bedding-corrected coordinates. In some samples, a low temperature component, which unblocks below 300–350 °C and conforms to the present geomagnetic field in geographic coordinates, is isolated in addition to a viscous magnetization removed below 120 °C. The component removed below 120 °C could occasionally correspond to a goethite component (Fig. 3f and g). At intermediate temperatures between 300–350 °C and up to 630–655 °C, there is little NRM unblocking (Fig. 3a and d) or a small component similar in direction to the ChRM is unblocked. (Fig. 3c and i).

It is interesting to observe the evolution of the intermediate temperature component across a polarity reversal. The normal to reverse transition N3-R4 at about 45 m above the section base (see below) is illustrated with five demagnetization diagrams (Fig. 3d–h). It can be observed that the intermediate temperature component reverses polarity at a lower stratigraphic level than the high-temperature component unblocked above 655–675 °C (note the stratigraphic level on each diagram from Fig. 3). This feature is particularly obvious for specimen R30-7A at 44.90 m (Fig 3g) but also in samples located tens of cm below (Fig. 3e and f). The presence of a magnetic component with the post-transition polarity at a level below the reversal boundary as defined by the ChRM component can suggest a delayed acquisition for the former component. Complex directional behaviour along the reversal boundaries of the Pliocene marine Trubi chalk Formation from southern Sicily has been explained by magnetite growth at different times at different levels in the sediment because of early diagenetic diffusion of Fe from the anoxic grey layers into suboxic/oxic zones where secondary magnetite would then form, resulting in delayed remanence acquisition (van Hoof and Langereis, 1991, see Dinarès-Turell and Dekkers, 1999, for a complete diagenetic model for the Trubi Formation). The sedimentary setting for the alluvial–fluvial red bed Buntsandstein succession is obviously different, but the magnetic fingerprint along a reversal may be comparable. We interpret the intermediate unblocking temperature component (below ~650 °C) as a chemical remanence (CRM) acquired by authigenic hematite grain growth in oxidizing conditions soon after deposition, and the high-temperature component as a detrital remanence (DRM) also carried by hematite. The presence of both components/mineral phases appears to be variable in the samples studied. The CRM component would therefore represent a delayed remanence able to record the post reversal polarity in the vicinity of a reversal boundary as observed in the case outlined above. These observations are consistent with petrological studies in the Buntsandstein red beds from the Iberian Ranges (Rey et al., 1996) (see below) and red beds elsewhere (e.g., Collinson, 1974), which despite the complexity of the Fe-oxide textural phases usually allow the assignment of coarse-grained hematite grains of detrital origin (specular hematite) to the high-temperature magnetic components, and fine-grained hematite (microcrystalline or pigmentary

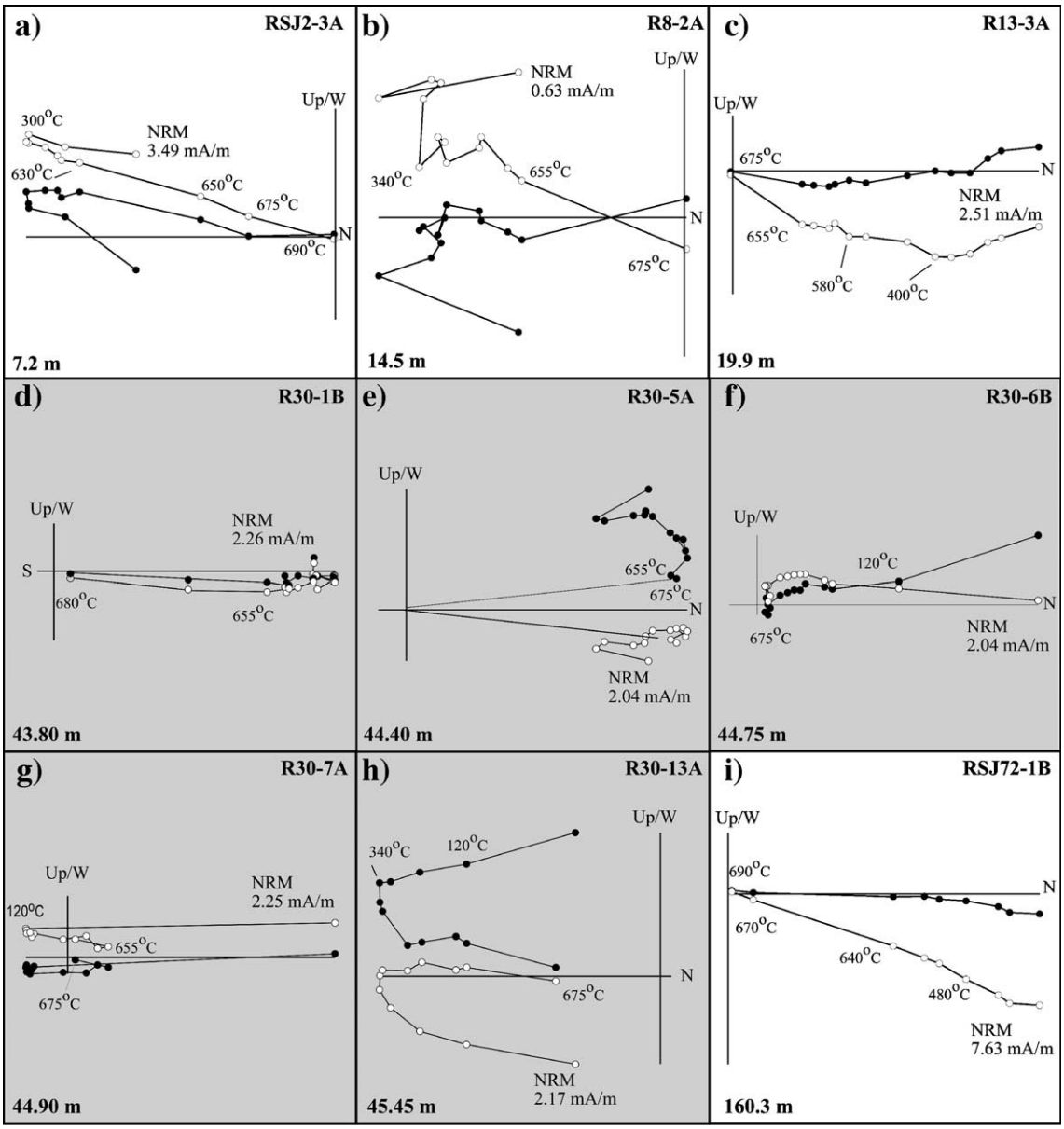


Fig. 3. Bedding-corrected orthogonal plots of thermal demagnetization data from representative specimens. Solid (open) symbols represent projections onto the horizontal (vertical) plane. The stratigraphic level, the NRM intensity and the temperature of some steps are indicated for each diagram. Diagrams (d) to (h) (shaded) correspond to the N3-R4 normal to reverse transition (see text for explanations).

hematite) to secondary overprints that unblock at intermediate temperatures. In the RSJ section, the secondary components appear to be an early diagenetic CRM although in other settings in the Iberian Range could be related to several phases of burial and uplift (Rey et al., 1996).

The reversal test of McFadden and McElhinny (1981) has been performed on the ChRM components in order to assess the antipodality of the normal and reverse populations (Fig. 4). This test classifies a 'positive' reversal test on the basis of the angle γ_c between the mean directions of the two sets of obser-

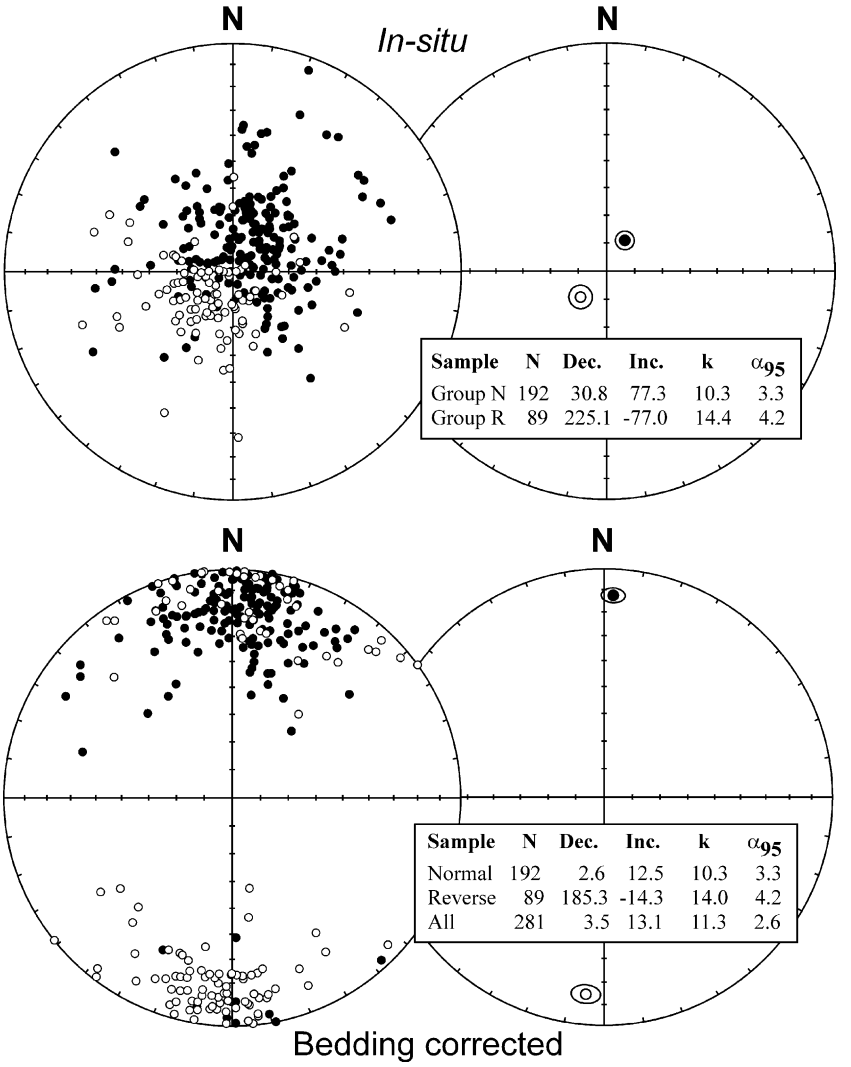


Fig. 4. Equal area projections of the ChRM directions before (in situ) and after bedding correction. The 95% confidence ellipse for the normal and reverse mean directions is indicated in the right projection. Statistical information is given (*N*, number of samples; Dec., declination; Inc., inclination: *k*, Fisher's precision parameter; α_{95}, radius of the 95% confidence cone).

vations at which the null hypothesis of a common mean direction would be rejected with 95% confidence (class 'A' if $\gamma_c \leq 5^\circ$ as 'B' if $5^\circ < \gamma_c \leq 10^\circ$, as 'C' if $10^\circ \leq \gamma_c \leq 20^\circ$, and 'indeterminate' if $\gamma_c > 20^\circ$). The ChRM data for the RSJ section pass the reversal test as class B.

The primary nature of the ChRM is supported by (1) the presence of a dual-polarity ChRM in addition to the low temperature present-day field overprint; (2) an unrealistic steep inclination before bedding correction (e.g., not compatible with any geomagnetic field direction for Iberia), and a shallow overall inclination after bedding correction consistent with the expected Early Triassic palaeoinclination (Osete et al., 1997; Van der Voo, 1993); (3) changes in polarity do not seem to be lithologically controlled.

The VGP latitude derived from the ChRM directions yields a succession of 5 magnetozones (3 normal and 2 reverse) that have been labeled from oldest to youngest as N1-R1-N2-R2-N3 (Fig. 5).

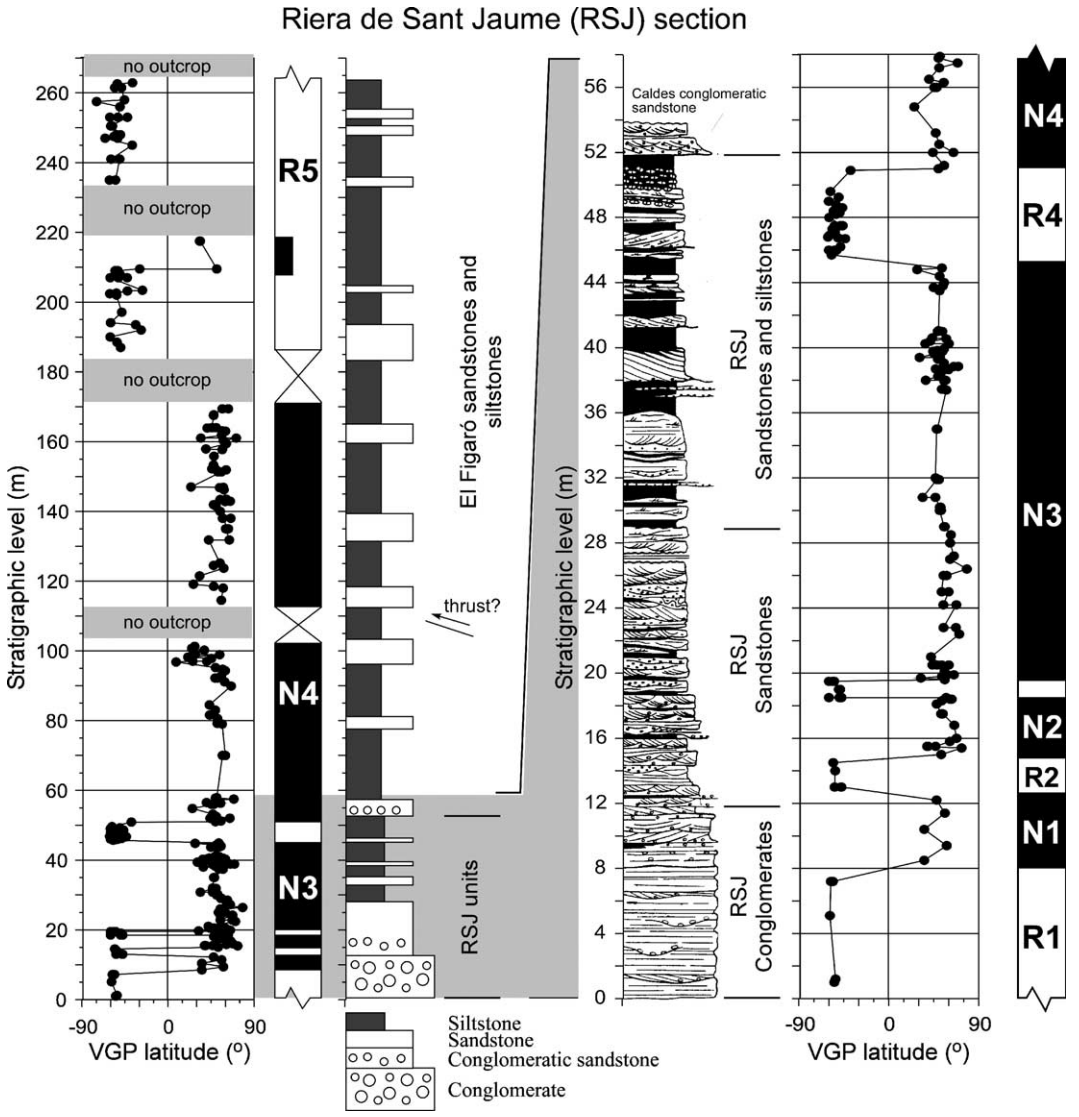


Fig. 5. Stratigraphic variation of the virtual geomagnetic pole (VGP) latitude (derived from ChRM directions) and interpreted magnetic polarity stratigraphy plotted on a lithologic log (modified from Marzo, 1980 and Calvet and Marzo, 1994). An enlargement of the lowermost 58 m of section comprising the Riera de Sant Jaume (RSJ) units is shown to the right.

5.2. The Iberian Range sections

The magnetostratigraphy of the RMS and TMMS Formations in the Molina de Aragón area along 5 sections (Rillo de Gallo, La Hoz del Gallo; Aragoncillo; Ermita de la Virgen del Buen Labrador and Arroyo de San Roman, Fig. 6) has been reported before (Rey et al., 1996). Detailed petrological observations and rock magnetic experiments where conducted in that study. Detrital hematite and ilmenohematite carry a primary magnetization in the form of a DRM or PDRM. It was inferred that a likely early CRM also contributed to the primary remanence. Goethite and several textural phases of authigenic hematite are responsible for a recent overprint in the form of a CRM. Acquisition of a PTRM during the uplift also contributed to the overprint. Despite the primary remanence that was slightly

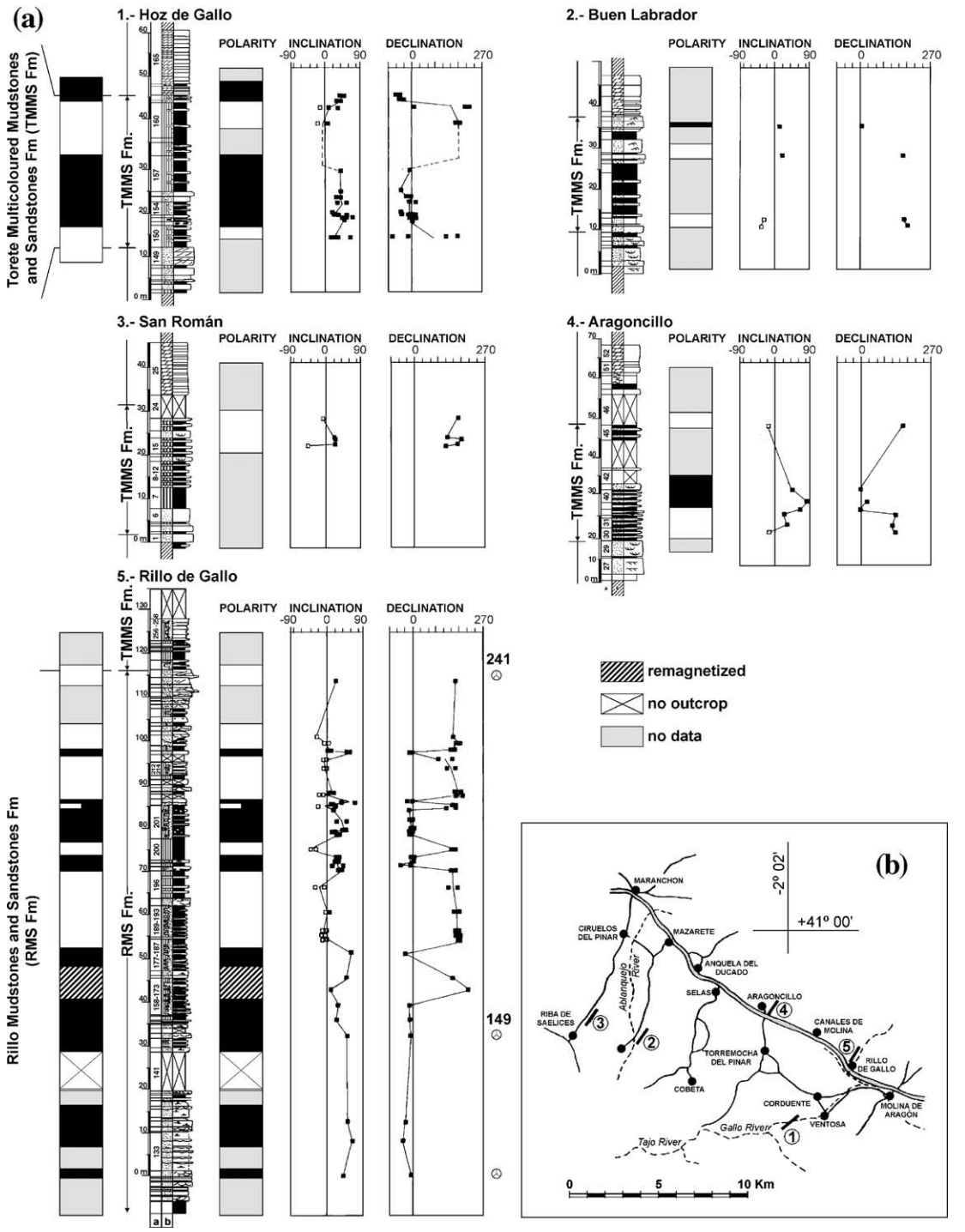


Fig. 6. (a) Correlation and composite magnetostratigraphic sequence for the Rillo Mudstones and Sandstones (RMS) and Torete Multicoloured Mudstones and Sandstones (TMMS) Formations in the Molina de Aragón area (after Rey et al., 1996). Lithological logs from Ramos (1979). (b) Location of the five studied sections in the framed area of Fig. 1.

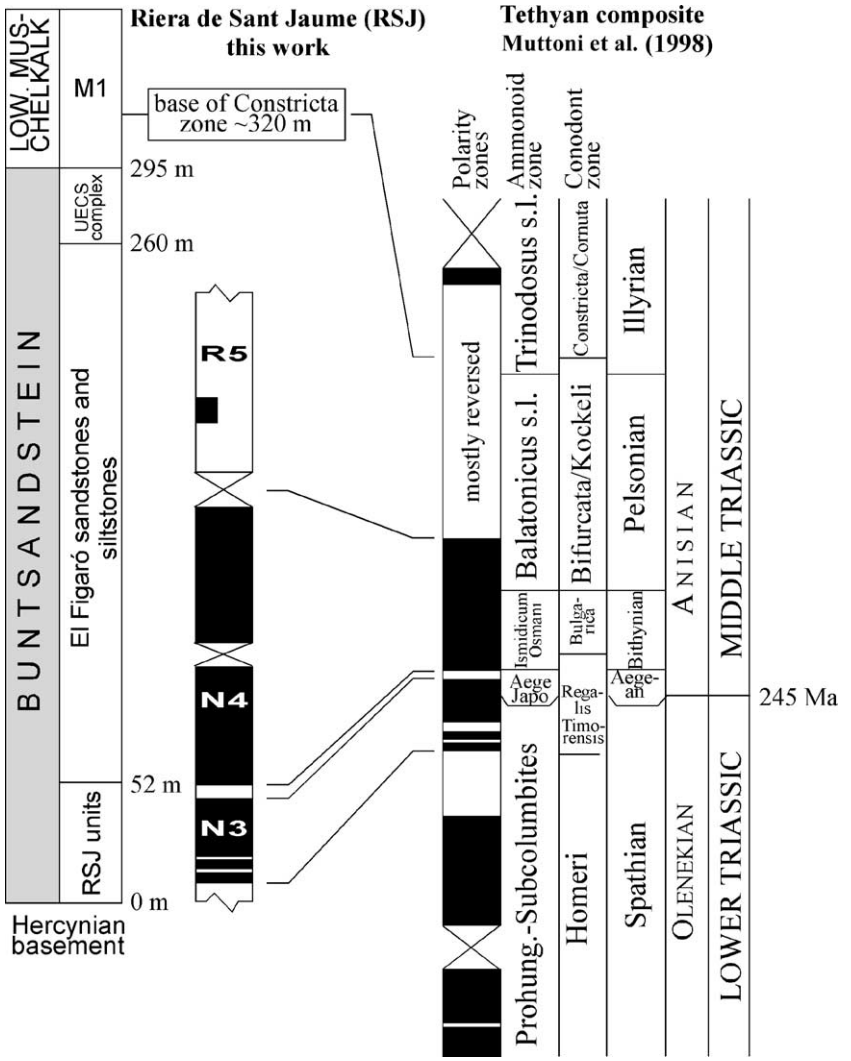


Fig. 7. Magnetostratigraphy of the Riera de Sant Jaume (RSJ) Buntsandstein section compared to biostratigraphically dated Tethyan composite section of Muttoni et al. (1998), which is based in this interval on data from Albania (Kçira composite, Nderlysacj sections) and Greece (Chios). Conodont biostratigraphic estimate (base of *Constricta* zone) is from Marquez-Aliaga et al. (2000). Numeric age value is from Gradstein et al. (2004).

biased by residual components of recent origin, a reliable Triassic magnetostratigraphy for Anisian–Ladinian times could be established. Fig. 6 illustrates the magnetozones defined at the different sections studied by Rey et al. (1996) and the synthetic polarity sequence.

6. Magnetostratigraphic correlation and discussion

After consideration of the biostratigraphic constraints outlined above, a straightforward correlation of the RSJ polarity sequence to the conodont-ammonoid-calibrated magnetostratigraphy from the Tethys

Fig. 8. Composite magnetostratigraphy of the Molina de Aragón Buntsandstein sections of Rey et al. (1996) compared to biostratigraphically dated Tethyan composite section of Muttoni et al. (1998). The recently revised location of the Anisian Ladinian boundary (Muttoni et al., 2004a,b) is considered. Age reinterpretation based on palynological assemblages proposed by Diez (2000).

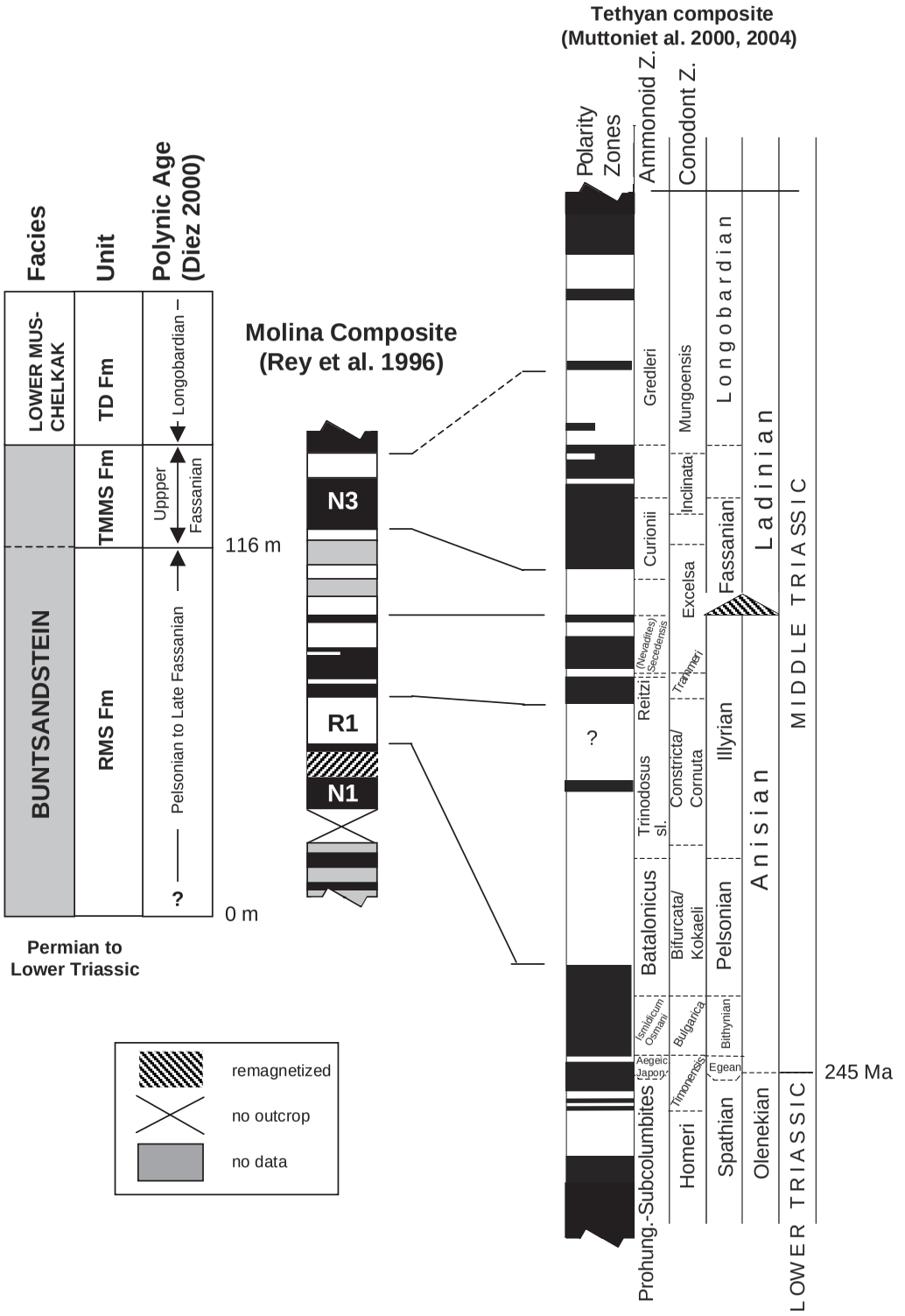

Facies
Unit
Polynic Age (Diez 2000)
LOWER MUS-CHELKAK
TD Fm
Longobardian
TMMS Fm
Uppper Fassanian
BUNTSANDSTEIN
RMS Fm
Pelsonian to Late Fassanian
?
116 m
0 m
Permian to Lower Triassic
Molina Composite (Rey et al. 1996)
N3
R1
N1
remagnetized
no outcrop
no data
Tethyan composite (Muttoniet al. 2000, 2004)
Polarity Zones
Ammonoid Z.
Conodont Z.
Prohung.-Subcolumbites
Homeri
Spathian
Olenekian
LOWER TRIASSIC
Aegeic
Japon.
Timorensis
Egean
Bithynian
Bulgarica
Ismidicum Osmani
Batalonicus
Bifurcata/ Kokaeli
Pelsonian
Trinodosus sl.
Constricta/ Cornuta
Illyrian
Anisian
MIDDLE TRIASSIC
Reitzi
(Nevadites) Secedensis
Trahyceri
Excelsa
Curionii
Fassanian
Gredleri
Inclinata
Mungoensis
Longobardian
Ladinian
245 Ma

Table 5
Late Paleozoic–Early Mesozoic selected paleomagnetic poles for Iberia

Rock unit, location	Age	Lat	Long	Pal.	K	α_{95}	Reference
Viar intrusives and red beds	Cu/Pl (263–296 Ma)	43	211	−1.2	110	6	Van der Voo (1969)
Buçaco red beds, N Portugal	Cu/Pl (268–291 Ma)	36	211	−7.5	332	7	Van der Voo (1969)
Buçaco red beds, N Portugal	Cu/Pl (268–291 Ma)	38	202	−8.4	206	9	Gomes et al. (2004)
Atienza andesites, NW Iberian Ranges	Cu/P (275–299 Ma)	36	203	−10.0	31	12	Van der Voo (1967)
Atienza andesites, NW Iberian Ranges	Cu/Pl (275–299 Ma)	42	208	−3.8	25	14	Hernando et al. (1980)
Atienza andesites, NW Iberian Ranges	Cu/Pl (275–299 Ma)	50	205	3.1	64	3	Osete et al. (1997)
Dikes, Catalan Coastal Range	Pl (268.286 Ma)	49	202	−3.6	83	9	Parés et al. (1988)
Red beds, RSJ section, Catalan Coastal Range	Tl–Tm (245–240 Ma)	55	172	5.0	11	3	This study
Red beds, Catalan Coastal Ranges	Tm (228–237 Ma)	51	181	1.1	205	6	Parés (1988)
Red beds, Molina de A., Iberian Ranges	Tm (230–240 Ma)	55	201	7.4	12	4	Rey et al. (1996), this study
Red beds, Alcázar de St. Juan, Southern Meseta	Tm–Tu (216–237 Ma)	63	178	13	47	3	Van der Voo (1967)
Red beds, various localities, Southern Meseta	Tm–Tu (216–237 Ma)	54	178	4.0	–	7	Parés and Dinarès-Turell (1994)
Red beds, Alcázar de St. Juan, Southern Meseta	Tm–Tu (216–237 Ma)	62	190	12.6	11	12	Osete et al. (1997)
Red beds, Alcaraz, outhern Meseta	Tm–Tu (216–237 Ma)	58	177	8.0	23	5	Osete et al. (1997)
Red beds, Ayllón, Iberian Ranges	Tu (216–228 Ma)	60	192	10.9	20	6	Osete et al. (1997)
Messejana–Plasencia dyke	Jl (201–205 Ma)	71	236	28.9	36	7	Schott et al. (1981)
Messejana–Plasencia dyk	Jl (201–205 Ma)	71	238	29.4	48	3	Palencia-Ortas (2004)

Abbreviations are C—Carboniferous, P—Permian, T—Triassic, J—Jurassic, l—lower, m—middle, u—upper; Lat/Long,—Latitude and Longitude of the paleomagnetic pole, Pal—Paleolatitude computed for a common location at 40°N, 4°W, k—Fisher's precision parameter, α_{95}—angle of the 95% confidence cone about the mean direction.

realm (Muttoni et al., 1998, 2000, 2004a) can be made (Fig. 7). Magnetozone R5 (that includes a potential small normal magnetozone) from the RSJ section is correlated to a mostly reversed zone in the early Anisian portion of the Tethyan composite and then the sequence of magnetozones N4 to R1 can be easily correlated to magnetozones spanning the early Anisian and the late Olenekian. Relative thickness of magnetozones among the Tethyan composite and the RSJ is comparable with the exception of magnetozone N4 from the RSJ section which appears relatively thick. Although this could be interpreted as a variation of sedimentation rate along the RSJ section (note however that the Tethyan composite is not time calibrated throughout), it could be also explained with a stratigraphic duplication related to a postulated thrust in the region (Berástegui et al., 1996) and discussed above. The polarity sequence retrieved at the RSJ section and the presence of distinct lithological units at the base of the section (the RSJ units, see Fig. 7) limit the eventual presence of a thrust within the El Figueró sandstones and siltstone unit and we have hypothetically placed it at about the 110 m level of the section studied in a covered interval (Fig. 5). The proposed magnetostratigraphic correlation indicates that the Olenekian (Scythian)–Anisian stage boundary (245 Ma) is located within magnetozone N3 in the Riera de Sant Jaume units. This is the first time that the Early Triassic is adequately identified within the Buntsandstein facies in eastern Iberia. However, it is likely that the Early Triassic is restricted to the Riera the Sant Jaume area given that the RSJ units have only been recognized there. For the rest of the Catalan Coastal Ranges (or at

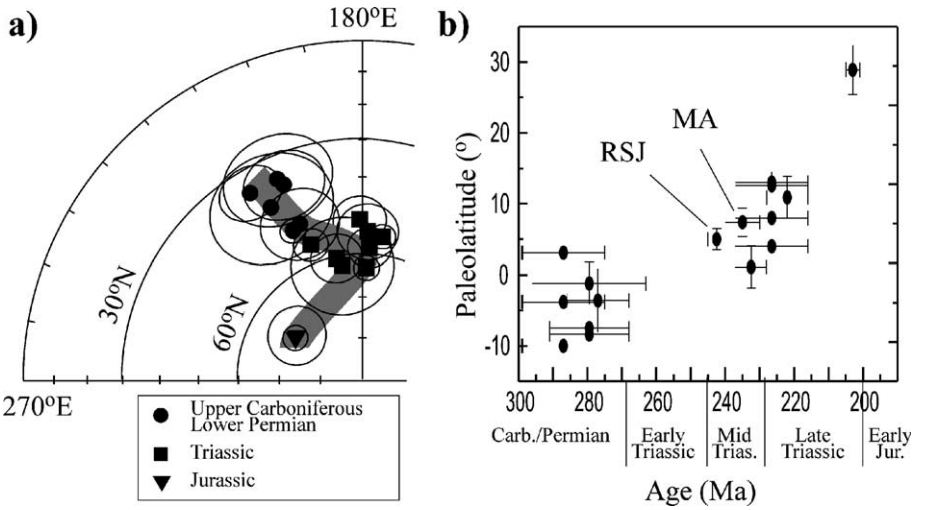


Fig. 9. Late Paleozoic to Early Mesozoic polar wander path (a) and paleolatitude (b) for Iberia (based on data from Table 5). The Riera de Sant Jaume section (RSJ) and the Molina de Aragón (MA) area data from this study are shown. The age scale is based on Gradstein et al. (2004).

least for the Montseny–Llobregat domain) the base of the Buntsandstein series corresponds to the El Figueró sandstones and siltstones units (the Caldes conglomeratic sandstone) (Calvet and Marzo, 1994), and therefore, it would be already early Anisian in age (see Figs. 5 and 7).

The composite magnetostratigraphy for the Molina de Aragón area (Fig 8) tightly constrained on the palynological age revision of RMS and TMMS units of Diez (2000) allows precise correlation with the Tethyan composite of Muttoni et al. (1998) (Fig. 8). It needs to be noted that the position of the Ladinian–Anisian boundary in this diagram has been located accordingly with the recent revision of Muttoni et al. (2004a) for these stages. Magnetozone R1 spans the Pelsonian Illyrian interval and the short normal interval in R3 marks the Anisian–Ladinian boundary. This correlation provides the basis to reinterpret the age of both formations, assigning a Pelsonian to early Fassanian age to the RMS Formation and a mostly late Fassanian age for the TMMS Formation, with unusual substage accuracy for red bed continental sequences. Equally important is the good agreement between the magnetostratigraphic zonation obtained in these fluviatile sequences that supports the marine data, and ultimately, their significance to tie palynological assemblages to the conodont and amonoid zones.

The Early/Middle Triassic paleopole for the Catalan Coastal Ranges is located at 55.1°N 172.4E (Dp=1.4, Dm=2.7), and the Middle/Late Triassic paleopole for the Iberian Ranges is 55°N 201E (Dp=1.7, Dm=3.1). We have compiled the paleomagnetic poles for the Late Paleozoic to Early Mesozoic portion of the Iberian apparent polar wander path (Table 5). The computed paleopoles for the Catalan Coastal Ranges (RSJ) and Iberian Ranges (MA) are compatible (Fig. 9) with the general trend of the Iberian apparent polar wander path which indicates a northward motion during the Triassic related to the general northward translation of Pangea.

7. Conclusions

We have presented new magnetostratigraphic data and reviewed available palynostratigraphic data for the Buntsandstein facies (Germanic Triassic) for eastern Iberia. The magnetostratigraphyies retrieved in the sections from the Catalan Coastal Ranges and Iberian Ranges can be collectively correlated to the conodont-ammonoid-calibrated magnetostratigraphy from the Tethys realm (Muttoni et al., 1998, 2000, 2004a) and allow the identification of the Olenekian–Anisian and Anisian–Ladinian boundaries in these continental red beds. Our chronostratigraphic framework corroborates the classic view that the

Middle Triassic marine transgression (Muschelkalk) develops from east to west implying progressively younger ages for the upper continental Buntsandstein facies in the same direction as demonstrated in this work (the top of the Buntsandstein is older in the Catalan Coastal Ranges than in the Iberian Ranges). Finally, the computed paleomagnetic pole positions are compatible and contribute to a better knowledge and documentation of the Iberian plate motion related to the Pangea kinematics in Triassic times.

Acknowledgements

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